process releases latent heat of condensation which further heats the air and forces the air to go further up.

Convectional precipitation is heavy but of short duration, highly localised and is associated with minimum amount of cloudiness. It occurs mainly during summer and is common over equatorial doldrums in the Congo basin, the Amazon basin and the islands of south-east Asia.

Statement 2: CORRECT—Cyclonic Rain:

Cyclonic Rainfall is convectional rainfall on a large scale. The precipitation in a tropical cyclone is of convectional type while that in a temperate cyclone is because of frontal activity.

54. **Orographic thunderstorm:** Forceful upliftment of warm moist air parcel when it passes over a mountain barrier creates cumulonimbus cloud causing heavy precipitation on the windward side.

    Orographic 'Cloud bursts' are common in Jammu and Kashmir, Cherrapunji and Mawsynram.

55. **Statement 1: INCORRECT—**Squall = a sudden violent gust of wind or localised storm, especially one bringing rain, snow, or sleet.

    Statement 2: INCORRECT—Torrent = a strong and fast-moving stream of water or other liquid.

56. **Wind Shear:**
    - Wind Shear = differences between wind speeds at different heights.
    - Tropical cyclones develop when the wind is uniform.
    - Because of weak vertical wind shear, cyclone formation processes are limited to latitude equator ward of the subtropical jet stream. [Jet streams]
    - In the temperate regions, wind shear is high due to westerlies and this inhibits convective cyclone formation.

57. **Statement 1: CORRECT**—Due to centripetal acceleration (centripetal force pulling towards the center is countered by an opposing force called centrifugal force), the air in the vortex is forced to form a region of calmness called an eye at the center of the cyclone. The inner surface of the vortex forms the eye wall, the most violent region of the cyclone.